

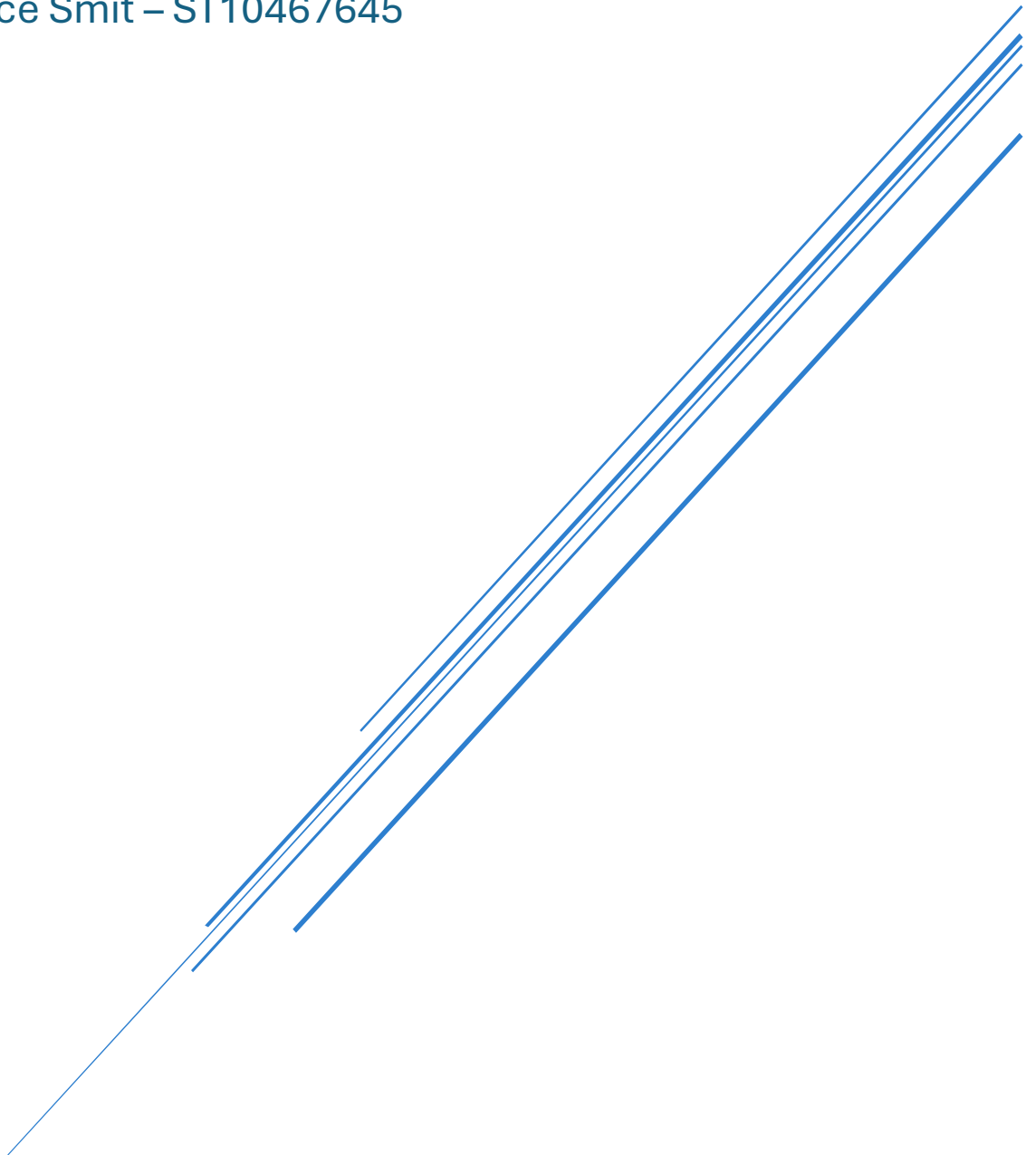
INFORMATION SYSTEMS: ICE

TASK 1 - REQUIREMENTS AND STAKEHOLDERS

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Identify and explain the functional requirements

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Scenario 1

The system needs to handle patient record management as its core function. This means staff must be able to create new patient profiles, update existing ones, and retrieve patient information quickly without having to sift through physical files. Every patient interaction, whether it's a first visit or a follow-up, should be captured digitally and stored in a structured way.

Appointment scheduling is another key functional requirement. The system must allow receptionists (and possibly patients) to book, reschedule, or cancel appointments. It should also track follow-up consultations and send reminders where necessary, replacing the manual register that receptionists currently manage.

Prescription management must also be built into the system. Doctors need to capture prescriptions electronically, and other authorised staff (like pharmacists or nurses) should be able to view them. This removes the dependency on paper-based prescriptions that can be lost or misread.

The system must support report generation; meaning it should be able to automatically pull together summaries, patient histories, and analytical reports. Right now, this is done manually and takes a lot of time, so automating it is a direct functional need.

Finally, there must be role-based access control. Different users (doctors, nurses, receptionists, administrators) interact with records in different ways, so the system must enforce who can view, edit, or delete certain information.

Scenario 2

The most central functional requirement is real-time traffic monitoring. The system must be able to collect live data from cameras, sensors, and other monitoring tools placed across the road network. This data forms the foundation for everything else the system does.

Linked to that is dynamic traffic signal control. Unlike the current fixed-schedule lights, the system must be able to automatically adjust signal timings based on actual traffic volumes at any given moment. For example, if one road is heavily congested, the system should extend the green light duration on that side to ease the flow.

The system also needs an AI-driven traffic analysis engine that can identify patterns, predict peak congestion periods, and recommend or automatically implement optimisations. This goes beyond just reacting as it should be able to anticipate problems before they escalate.

Route update and commuter notification is another functional requirement. The system must push real-time updates to drivers through platforms like a mobile app, GPS integration, or electronic road signs. Drivers should be able to see congested areas and get suggested alternative routes.

Incident detection and alerting is also required. If there's an accident or an unusual blockage, the system should detect it quickly and alert both traffic authorities and commuters so that responses can be coordinated faster.

Lastly, the system must have a centralised dashboard for traffic authorities. This is where operators can monitor the entire city's traffic in one place, override automated decisions if needed, and generate reports on traffic patterns over time.

Identify and explain the non-functional requirements in the context of the business needs

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Scenario 1:

Non-functional requirements refer to how a system should behave rather than what it does. In the case of healthcare, these requirements are critical for safety and trust. The non-functional requirements for Scenario 1 are:

Data Integrity: To fix the inconsistencies mentioned in the manual system, the digital records need to be highly accurate and protected against any accidental deletion or corruption during updates.

Usability: The new interface needs to be intuitive to minimize the learning curve and prevent data entry errors.

Reliability and Availability: The system needs to have high uptime due to the medical dependency on these records. Doctors will require constant access to the system so that they can view patient history during emergencies without any system downtime.

Security: Since physical folders are being moved into a digital database, the system must ensure patient confidentiality. This can be achieved through encryption and role-based access control to prevent unauthorized data breaches.

Scenario 2:

When it comes to the city-wide traffic system, the focus shifts towards real-time processing and operating at a massive scale. The non-functional requirements for Scenario 2 would be:

Scalability: The system should be able to manage data from multiple sensors, cameras, and GPS units across the city simultaneously without crashing.

Accuracy: The AI's "intelligent coordination" must be highly accurate in identifying vehicle density to ensure that signal timings improve traffic flow rather than hinder it.

Interoperability: The system would need to communicate with different types of hardware installations and third-party navigation applications.

Identify the different stakeholders across the different dimensions – Internal and external, operational and executive.

Scenario 1

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Stakeholders

Internal Stakeholders

Operational level:

- Receptionists – They manage appointments and the front desk register daily. They'll be the most frequent users of the scheduling and record retrieval functions.
- Doctors/Physicians – They need to access patient histories, write prescriptions, and record diagnoses. Therefore, they'll interact with the system at a clinical level.
- Nurses – They update patient progress, administer prescriptions, and record follow-up observations.
- Pharmacists – They rely on prescription data from the system to dispense medication correctly.

Executive level:

- Hospital Administrator/Management – They oversee the overall operations of the hospital and will use the reporting and analytics features to make decisions about staffing, resource allocation, and compliance.
- IT Department – Responsible for maintaining, securing, and updating the system once it is deployed.

External Stakeholders

Operational level:

- Patients – The primary beneficiaries of the system. They benefit from faster service, accurate records, and better-managed appointments.

Executive level:

- Government/Regulatory Bodies – Since this is a government hospital, bodies that oversee healthcare standards and data compliance (e.g., health departments or data protection authorities) have an interest in how patient data is managed and secured.
- Medical Aid/Insurance Providers – They may interact with the system when processing claims that require verified patient records or treatment histories.

Scenario 2

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For Scenario 2, the operational stakeholders would be traffic controllers, maintenance crews, and drivers. This is because traffic controllers will monitor the AI system, maintenance crews will repair and maintain the hardware, and drivers will rely on real-time updates for their daily commutes.

The executive stakeholders would be the City Council, urban planners, and environmental agencies. This is because their focus will be on reducing environmental pollution and managing the infrastructure budget.